



Inclusive design: embracing diversity

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Introduction

User-Centered Design, as the name suggests, revolves around the user. It takes into account his goals and his expectations as well as his struggles, both known and possibly discoverable. Every designer must carefully evaluate the group of people for whom he is designing and, to do so, research is fundamental.

However, while trying to identify the group of target users we are at risk of encountering many possible mistakes: one of them is concentrating exclusively on common, typical or even “normal” users, while ignoring other categories that will be excluded by default.

This is a major disadvantage both for the user, who hasn't been considered during the design process, and for the client, whom product will have a limited audience.

User Experience doesn't focus only on the usability but also on the pleasurability of the design. So even if a user can practically and successfully use the product or service offered, we can't exclude the possibility that he may not enjoy it.

In this paper I'll address the concept of inclusivity, its central role in the design of products and services, who is the user and how to avoid exclusion, the responsibility of the designer, how to apply inclusivity in the design process and its similarities and differences with accessibility.

Inclusive design

Inclusive design is a design philosophy that aims to create solutions that can benefit as many people as possible. It takes a step away from the concept of “one-size-fits-all”. Inclusive design explores, embraces and considers diversity, in all its forms.

It allows a greater number of people to engage with products and services and to benefit from what they have to offer.

It enables everyone to participate equally, confidently and independently in everyday activities and creates new opportunities to deploy creative and problem-solving skills.

Inclusive design is also about acknowledging the differences and offering choices and opportunities. It provides flexibility and enjoyment, it accommodates all people, regardless of their age, gender, ethnicity, circumstances, experience, abilities and disabilities. Every barrier becomes an opportunity for innovation and creativity.

Why is it important and what's the role of the designer?

Design can have a great impact. It can help reaching a wider audience, extending the market, as well as having great consequences on society, offering the same opportunities. Inclusive design also reflects how people really are and helps with the process of normalization of diversity.

It's important as designers to use our creativity and lateral thinking to find innovative and individual solutions, designing for real people in all their variability, but inclusivity is everyone's responsibility. Every environment, physical and virtual should be designed following this philosophy, to lower barriers to participation in society.

What's exclusion and how does it happen?

The user, by definition, is someone who uses a product, machine, or service.

When designers do the research with users they tend to focus on the discovery of similarities and patterns, useful to create user personas. This process goes against what inclusive design is about.

“Learning how to recognize exclusion is the first step in making experiences exclusive. When we let new perspective supersede our own, we shift toward more inclusive solutions. It's a skill that we can develop with practice”

cit. Kat Holmes
Author, Director of UX Design, Google

The first mistake is made while defying the user personas. Their creation is a valid tool that helps to focus on the target of the product or service throughout its design,

but they can also be a dangerous element of shallowness. The paradox is that through the user persona we are forced into a simplified view of complex, different and diverse beings: people. Piotr Źrołka, accessibility expert, suggests we change the way we see our users. User personas obstruct our view of the most important facets of people (needs, limits, desires) with sterile data, like demographics. The solution is to not linger on appearances, but to work hard in the research phase. Therefore, it's crucial to include people in the process, allowing the creation of innovative and creative solutions, as they can provide the designer with a wide range of perspectives.

Inclusive design puts people in the center from the very start of the process, and those fresh, diverse perspectives are the key to true insight. Microsoft Design suggests using the Persona Spectrum as a tool to help foster empathy.

When designing, we should ask ourselves “who are we excluding?” and “what abilities are necessary to use our design?”. These questions can help stimulate empathy and reflection.

Moreover, empathy is built through listening. A designer should be very careful not to fall into biases. Instead, they should always be brave enough to go against themselves, question their decisions and avoid being over-confident when research does not support the design decisions made.

Lastly, another important strategy is to always try to offer alternatives to the user and let him explore and experience the product or service freely (or at least in a personalized way).



Accessibility vs Inclusivity

Accessibility is about granting the same experience opportunities to people affected by a disability, removing all barriers that prevent them from interacting and accessing a product, service or interface.

Accessibility is an attribute, while inclusive design is a method. Ideally, accessibility and inclusive design work together to make ensure that experiences are truly usable, pleasurable and open to all.

As far as accessibility, in the WCAG 2.1 (Web Content Accessibility Guidelines), written by the W3C (World Wide Web Consortium), are listed rules to design for accessibility. Here are some of them.

Ensure contrast is adequate: a light background with a dark text is the basic combination to provide good legibility. Broadly speaking, the goal is a contrast ratio of 4.5:1, while a ratio of 3:1 is necessary when using fonts bigger than 18 pt. It's possible to test the design by visiting www.accessible-colors.com. This rule should also be applied to images, not only texts.

Use blank spaces: leave a blank space between different parts of the page is really helpful to differentiate them and to give a more organized view of the page, which also helps in processing the information.

Take into account readability: apart from legibility, readability must also be considered. It's about the content of the text and its ease of understanding. Therefore, it pays to employ a clear, fluent and simple speech. This is essential for dyslexic users, mostly. You can evaluate your text readability using the READ-IT demo.

Incorporate transcriptions in pictures and videos: audio and video products should always have a text alternative, as they can be useful to visually impaired users using screen readers. A transcription can also be useful to users with a short attention span or with difficulties in understanding the spoken word, for example.

Be careful with light effects: it is advisable to avoid designing products that can cause epileptic seizures, due to light-sensitive epilepsy, like fast blinking lights or fast changing pictures. If they can't be avoided a warning should be included before the offending effect.

Always give a feedback: the user must also be sure that his request is being processed: as such, it's important to use more than one sensory feedback, like a text and a sound.

Account for people using exclusively a keyboard: not every user is going to use both mouse and keyboard. For example, the selected item should be highlighted even if not selected through a mouse.

Do not use (only) colour to mark, divide and differentiate items: while browsing, it's better to highlight the displayed page underscoring its name or making it bold. Using only colour is not enough to emphasize an item, like an error message. It's also advised to write it bigger and/or bold. This is necessary for colorblind people. Online there are many tools to verify if a colour palette is suitable.

Attach written descriptions to pictures: always add an alternative text to pictures, as some people use assistive technologies, later discussed, that can read the description added to an image. However, descriptions shouldn't be too long to prevent slowing the experience.

Be careful when selecting a font: the font must be readable, of an adequate size and spacing. Usually, the easier-to-read fonts are the sans-serif, but it's not an iron rule. Some of the most popular are Arial, Helvetica, Montserrat, Open Sans and Lato.

What is a disability

An interesting definition of disability is given by the Social Model of Disability, which differs from the Medical Model. According to it, a disability is the result of the interaction between the physical or sensorial or cognitive or mental individual limitation and the context of life.

In other words, a person is disabled because the context makes him or her such. This definition puts society and, more specifically, the designer in a position of responsibility. Nobody would feel disabled if products and services were designed without barriers. Also, the WHO changed the definition of disability, from a health problem to the "consequence of the interaction between features of a person's body and features of the society in which he or she lives". This results in mismatched interactions.

A disability can be permanent, temporary or situational.

1. Vision impairment: blindness, hypovision, color-blindness
2. Motion impairment: total or partial absence or immobility of limbs, slowness, shakiness and muscular pain.
3. Hearing impairment: of every level, to complete deafness.
4. Cognitive impairment: dyslexia, attention, comprehension and memory difficulties.

Following the definition of temporary impairment, we can imagine other disabilities caused by context.

1. Vision impairment: sudden flash.
2. Motion impairment: broken arm or holding a baby.
3. Hearing impairment: being in a loud place.
4. Cognitive impairment: foreign language

Assistive technologies

They are IT tools used to grant disabled people access and use to hardware and software. They are interfaces that can be installed on a standard system, like a computer or a smartphone, to make it accessible.

Here are some examples:

- alternative browsers
- eye-trackers
- personalized keyboards
- ergonomic mouse
- pression sensors
- screen readers

Solve for one, extend to many

In the second episode of the first season of “Wireframe” podcast Khoi Vinh e Isabella Kulkarni tell us how inclusivity was born.

The first tv show to use captions was Julia Child’s, a chef that presented French dishes to americans. Captions can be useful to many people: those with hearing impairment, foreigners, but also to those who simply like to read and turn off the volume. However, captions were designed specifically for the audience with hearing impairment.

The use of open-captions was not liked by everyone: they were not removable so even those without hearing problems were forced to see them. Many found them annoying and limiting, as they were covering the scene.

The mistake was that trying to find an accessible solution to those with a disability, the designers forgot about those that don't have it or that simply won't like it. In trying to remove a limit, they created one.

The problem was then solved with closed-captions: optional captions that the user could active whenever he wanted.

A recent example are audiobooks. They help those with a visual or motion disability but at the same time they help also those who want to read while doing something else.

This is inclusivity: to design for diversity, not for limits.

In her TED Talk Elise Roy explains that she gradually lost the ability to hear and she asked “what if we start to design for disabilities first? When we do so, we find new solutions that are inclusive and most of the times even better”.

In conclusion, it's necessary to start designing something accessible, with an inclusive mind.

P.A.C.T.

A good approach to analyze the relationship between people, activity, context and technology to design a good interface is to apply PACT. It's usually the result of a group brainstorming, with the objective of identifying the needs for the designing of interactive systems.

People do activities in a specific context with technologies:

1. People: we must think of physical differences (height, weight, disabilities,...), psychological ones (language, memory, desires,...), social ones (motivations, hobbies,...) and mental models (built on expectations and experiences);
2. Activities that people want to do: there are many attributes: time (regularity and duration), complexity, safety, pressure, continuity and the nature of the practice;
3. Context where the activities happen: it includes the material place (location, noise, crowds,...), the social context and the organizational one;
4. Technology: the hardware and software components. We must consider input, output and communication.

How can all of this be useful in regard to accessibility? Thinking of people instead of persona and of the context where the product or system will be used, together with the way it will be employed, can be useful to assess new possibilities and discover obstacles not considered before.

References

Youtube Video: Inclusive Design Is a Welcome Mat, NNgroup.

<https://www.youtube.com/watch?v=2pBILUE49hI>

Microsoft Design, Inclusive Design.

<https://www.microsoft.com/design/inclusive/>

The principles of inclusive design, CABE (Commission for Architecture and the Built Environment), 11th may 2006.

<https://www.designcouncil.org.uk/sites/default/files/asset/document/the-principles-of-inclusive-design.pdf>

Web Content Accessibility Guidelines (WCAG) 2.1.

<https://www.w3.org/TR/WCAG21/>

Webinar: "11 ways to start your inclusive, accessible design toolkit", UXPin, with Piotr Źrółka, 28th january 2021.

<https://www.youtube.com/watch?v=1-IJzpTJYMU>

Podcast "Wireframe" by Adobe: "Inclusivity is a recipe for Good Design, Season 1, Ep. 2, 20th november 2018, with Khoi Vinh and Isabella Kulkarni.

<https://xd.adobe.com/ideas/perspectives/wireframe-podcast/inclusivity-is-a-recipe-for-good-design-episode-7/>

Youtube: TED, "When we design for disability, we all benefit", by Elise Roy.

<https://www.youtube.com/watch?v=g2m97gPI70I>

UX Planet: "Usability for designers, P-A-C-T framework", written by Muditha Batagoda, 27th january 2018.

Firth A. (2019), Practical Web Inclusion and Accessibility: a Comprehensive Guide to Access Needs, Apress.